The MAX7219 dot-matrix LED module has a serial input common-cathode driver that interfaces microcontrollers to LED matrices. All common microcontrollers can be connected to this module by using a 4-wire serial interface. Each output can be addressed without refreshing the entire display.

This module requires only three input/output lines for driving the display, making it ideal for microcontroller projects. Only one external register is used to set the segment current of each LED.

HC-05 Bluetooth is used for applications like wireless headset, game controllers, wireless mouse, wireless keyboard, besides some other consumer applications and student projects. It has a range up to 100 metres, which depends on the transmitter and receiver used, the atmosphere, and geographic as well as urban conditions.

HC-05 uses IEEE 802.15.1 standardised protocol through which one can build wireless personal area network (PAN). It uses frequency-hopping spread spectrum radio technology to send data over air and serial communication to communicate with devices. It communicates with the microcontroller using serial port with a universal baud rate of 9600 bps.

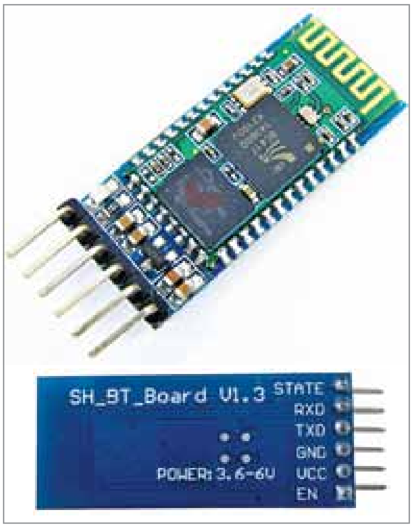

Fig. 3: HC-05 Bluetooth module

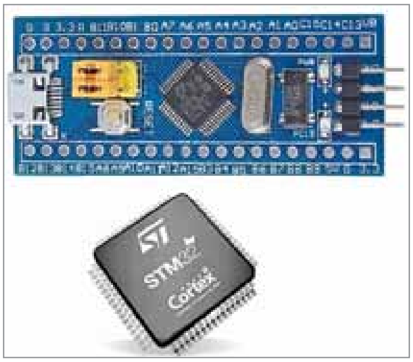

Fig. 4: STM32F103C8T6 Blue Pill board

TABLE 2
SPECIFICATIONS OF THE STM32 (BLUE PILL)

Name	Description
Architecture	32-bit ARM Cortex M3
Operating voltage	2.7V to 3.6V
CPU frequency	72MHz
Number of GPIO pins	37
Number of PWM pins	12
Analog input pins	10 (12-bit)
USART peripherals	3
I2C peripherals	2
SPI peripherals	2
Can 2.0 peripheral	1
Timers	3 (16-bit) (1 PWM)
Flash memory	64Kb
RAM	20Kb

HC-05 is a Bluetooth module designed for wireless communication. The module can be used in a master or slave configuration. Fig. 3 shows how it looks.

Fig. 4 shows the Blue Pill controller, which costs very less compared to an Arduino board. The hardware is open source. The microcontroller in it is the STM32F103C8T6 from STMicroelectronics.

The board also has an 8MHz crystal oscillator and a 32kHz crystal oscillator, which can be used to drive the internal real-time clock (RTC). Because of this, the microcontroller can operate in deep-sleep mode, making it ideal for battery operated applications.

The ARM Cortex M3 STM-32F103C8 microcontroller is used in the Blue Pill board. The microcontroller nomenclature STM32F103C8T6 has a meaning behind it:

- STM stands for the manufacturer's name STMicroelectronics
- 32 stands for 32-bit ARM architecture
- F103 indicates the architecture ARM Cortex M3
- C stands for 48-pin structure
- 8 stands for the 64kB Flash memory
- T indicates package type is LQFP
- 6 for operating temperature of -40°C to +85°C

Circuit and working

Circuit diagram of the LED dot-matrix scrolling display circuit can be divided into two parts—the regulated power supply and the control circuit.

The regulated power supply is shown in Fig. 5. It has a full-wave rectifier built around 1N4007 rectifier diodes (D1, D2) and a 5V voltage regulator (IC1). Its centre-tapped transformer X1 reduces 230V AC mains to 12V-0-12V, 500mA. The capacitors C1 and C2 filter the ripples. The regulated 5V (Vcc) output from

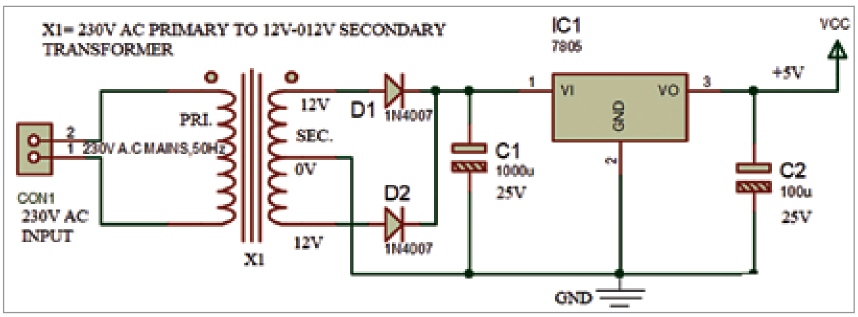

Fig. 5: Regulated power supply circuit